

W5 Exercises



The last task is a bit challenging. I **highly recommend** trying it on your own first and checking the notes if you run into any difficulty.

Task 0

- Set the seed to be `2026` for each task when needed!

Task 1: Control Flow

- Print out the characters `"NTU"` for four times, and then print out the characters `"ECON department"` for six times.
 - Hint:** there are multiple ways to accomplish this, but I encourage you to try to use a single `for` loop + an `if-else` statement to complete the task

Task 2: Binomial Random Sample

- Draw `400` observations from binomial distribution (total `1000` trials and `0.5` probability)
- Compute the mean of the observations
- Use the random sample you just generated. Create a dataframe to store the drawn values and draw a plot to depict the distribution
 - Hint:** Be aware of whether the plot you choose is appropriate!

Task 3: Certification Exam



A certification exam is organized for 10 students: Alice, Bob, Charlie, Diana, Ethan, Fiona, George, Hannah, Ivan, and Julia. Each student participates in 10 exams, named `exam_1`, `exam_2`, ..., `exam_10`. If the students pass certain amount of tests, they can claim the certificate.

- Check if you have set the seed `2026` (otherwise you will always get different results)
- Create a vector `students` that contains `10` students' name
- Create a matrix `scores` that use `rnorm()` to generate students' score in `10` exams (average = `60`, standard deviation = `15`). Set the column name of the matrix to be `exam_1`, `exam_2`, ..., `exam_10`
 - Hint:** use `replicate()` to create the matrix and create a vector `exam_names` for changing column names

- Create a dataframe `stu_scores` that combines `students` and `scores`
 - **Hint:** use `data.frame(name = students, scores)`
- Check what's in `stu_scores` using `View(stu_scores)` and take a deep breath...

💡 For later steps, you need to fully understand what you are doing with your dataframe! Try to complete them or discuss with TAs!

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- Create a function `certificate_winner` containing three parameters: `data`, `passing_score`, `min_passed` such that a student
 - passes the test if his/her score of that exam is greater than or equal `passing_score`
 - wins the certificate if the number of passing test is greater than or equal `min_passed`

Finally, the function should return the students who wins the certificate.

- **Hint:** set a counter `pass_count` to count the number of exams a student passes and create a vector `passed_students` to save the student name for those who can claim the certificate
- Answer the following questions:
 - Who are the certificate winner if `passing_score` is 60, and `min_passed` is 5?
 - Who are the certificate winner if `passing_score` is 70, and `min_passed` is 5?

💡 Congratulations on completing the very difficult task!

If you're seeing this because you decided to give up, check the answer code and try to understand the logic! It will be helpful for data cleaning and data manipulation!



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